

1. A waveform $\phi(t)$ with similarity function $R_{\phi\phi}(\tau) = \text{sinc}(2\tau)$ is used to transmit information as

$$\sum_{i=-\infty}^{\infty} X_i \phi(t - iT),$$

where the X_i are symbols from an M -QAM constellation and T is the symbol period. The maximum rate without inter-symbol interference is

- (a) $2 \log_2 M$ bits per second. *
- (b) $\frac{\log_2 M}{4}$ bits per second.
- (c) $\frac{\log_2 M}{2}$ bits per second.
- (d) $\log_2 M$ bits per second.

Solution: The shifted waveforms $\phi(t - iT)$ are free of inter-symbol interference when they are orthogonal, that is, when the similarity function $R_{\phi\phi}(\tau) = \int_{-\infty}^{\infty} \phi(t) \phi^*(t - \tau) dt$ vanishes at every nonzero multiple of the symbol period, $R_{\phi\phi}(kT) = 0$ for $k = \pm 1, \pm 2, \dots$. Since $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$ vanishes at every nonzero integer x , the function $\text{sinc}(2\tau)$ vanishes at $\tau = \pm \frac{1}{2}, \pm 1, \pm \frac{3}{2}, \dots$. The smallest admissible symbol period is the first zero, $T = \frac{1}{2}$, and every multiple of it is then a zero as well. The symbol rate is $R_s = \frac{1}{T} = 2$ symbols per second, and since an M -QAM symbol carries $\log_2 M$ bits, the maximum bit rate is $R_b = 2 \log_2 M$ bits per second. The same value follows from the frequency domain: $\text{sinc}(2\tau)$ is the inverse transform of a rectangle on $[-1, 1]$, so $\phi(t)$ has bandwidth $W = 1$ and $R_s = 2W = 2$.

2. For some $a, b > 0$ with $a \neq b$, a modulation system maps two equiprobable independent bits (D_1, D_2) to the symbol

$$X = \begin{cases} a + jb & (D_1, D_2) = (0, 0) \\ a - jb & (D_1, D_2) = (0, 1) \\ jb & (D_1, D_2) = (1, 0) \\ a & (D_1, D_2) = (1, 1). \end{cases}$$

What is the average energy of the symbol X ?

- (a) $\frac{3a^2 + 3b^2}{4}$ *
- (b) $a^2 + b^2$
- (c) $2(a^2 + b^2)$
- (d) ab

Solution: The energy of a symbol is its squared modulus $|X|^2$, and the average energy is its expectation over the bits, $E_s = E(|X|^2)$. The four symbols have energies $|a + jb|^2 = a^2 + b^2$, $|a - jb|^2 = a^2 + b^2$, $|jb|^2 = b^2$ and $|a|^2 = a^2$. Since the two bits are independent and equiprobable, each pair (D_1, D_2) , and

hence each symbol, has probability $\frac{1}{4}$, so that

$$E_s = \frac{1}{4}(2(a^2 + b^2) + b^2 + a^2) = \frac{3a^2 + 3b^2}{4}.$$

The value $a^2 + b^2$ would be the energy if all four symbols were placed at the same distance from the origin, which is not the case here.

3. What is the most likely output of the following MATLAB code?

```
n = 100000;
aux = rand(1,n);
xx(aux > 0.25) = 1j;
xx(aux < 0.25) = 3;
1/n*sum(xx.*conj(xx))
```

- (a) 3 *
- (b) 5
- (c) 2
- (d) 7

Solution: The vector `aux` contains n independent uniform numbers in $(0, 1)$, and the two assignments build a vector `xx` whose entries are j when the uniform number exceeds $\frac{1}{4}$ and 3 otherwise. Hence `xx` is a sequence of n independent symbols X_i from the constellation $\{j, 3\}$, with $\Pr(X = j) = \frac{3}{4}$ and $\Pr(X = 3) = \frac{1}{4}$. The product `xx.*conj(xx)` is the squared modulus of each entry, that is, the energy of each symbol, and the last line averages these energies. By the law of large numbers, for large n the average of independent realizations approaches the expectation, so

$$\frac{1}{n} \sum_{i=1}^n |X_i|^2 \approx E(|X|^2) = \frac{3}{4}|j|^2 + \frac{1}{4}|3|^2 = \frac{3}{4} + \frac{9}{4} = 3.$$

The output is the average symbol energy of the constellation, close to 3.

4. A channel with input $X \in \{0, 1\}$ and output $Y \in \{0, 1\}$ has transition probabilities $W(1|0) = \frac{1}{2}$ and $W(0|1) = 0$. What is the mutual information $I(X; Y)$ when the input bits have probabilities $\Pr(X = 0) = 1 - q$ and $\Pr(X = 1) = q$? Here $h_2(p)$ denotes the binary entropy function.

- (a) $1 - q + h_2\left(\frac{1-q}{2}\right)$
- (b) $q - 1 + h_2\left(\frac{1+q}{2}\right)$ *
- (c) $q + h_2\left(\frac{1-q}{2}\right)$
- (d) $-q + h_2\left(\frac{1+q}{2}\right)$

Solution: We use $I(X; Y) = H(Y) - H(Y|X)$. When $X = 0$ the output is equiprobable, so $H(Y|X = 0) = 1$. When $X = 1$ the output is always 1, since

$W(0|1) = 0$, so $H(Y|X = 1) = 0$. Averaging over the input, $H(Y|X) = (1 - q) \cdot 1 + q \cdot 0 = 1 - q$. The output takes the value 1 with probability

$$\begin{aligned}\Pr(Y = 1) &= (1 - q)W(1|0) + qW(1|1) \\ &= \frac{1 - q}{2} + q = \frac{1 + q}{2},\end{aligned}$$

so $H(Y) = h_2(\frac{1+q}{2})$ and $I(X; Y) = h_2(\frac{1+q}{2}) - (1 - q)$. As a check, for $q = 1$ the input is deterministic and the mutual information is $h_2(1) - 0 = 0$. For $q = 0$ it is $h_2(\frac{1}{2}) - 1 = 0$ as well, since a deterministic input carries no information.

5. We want to transmit reliably over a complex AWGN channel with bandwidth $W = 2$ and noise spectral density $N_0 = \frac{1}{4}$ at a rate of $R = 4$ bits per second. This is possible if and only if the power P satisfies

- (a) $P < \frac{3}{2}$
- (b) $P > \frac{3}{2}$ *
- (c) $P > \frac{1}{2}$
- (d) $P < \frac{1}{2}$

Solution: The condition $R < C$ becomes $4 < 2 \log_2(1 + \frac{P}{N_0 W}) = 2 \log_2(1 + 2P)$, since $N_0 W = \frac{1}{2}$. Dividing by 2 and using that the logarithm is increasing, $\log_2(1 + 2P) > 2$ holds when $1 + 2P > 4$, that is, $P > \frac{3}{2}$. The boundary value gives exactly 4 bits per second: $2 \log_2(1 + 3) = 4$. Note the roles of the two parameters: the condition only involves the ratio P/N_0 , here required to exceed 6.

6. A coded-modulation system sends one information bit D as the pair of symbols $\mathbf{x}(D) = (X_1, X_2)$ with $X_1 = X_2 = \sqrt{E_s} D$, over the complex AWGN channel with noise variance $\sigma^2 = 1$. What is the set of all received pairs $\mathbf{y} = (y_1, y_2)$ for which the two likelihoods coincide, $W^2(\mathbf{y}|\mathbf{x}(0)) = W^2(\mathbf{y}|\mathbf{x}(1))$?

- (a) $\text{Re}(y_1) = \text{Re}(y_2) = 0$
- (b) $\text{Re}(y_1) = \text{Re}(y_2) = \frac{\sqrt{E_s}}{2}$
- (c) $\text{Re}(y_1) + \text{Re}(y_2) = 0$
- (d) $\text{Re}(y_1) + \text{Re}(y_2) = \sqrt{E_s}$ *

Solution: The channel is memoryless, so the likelihood of the pair is the product of the likelihoods of its two symbols,

$$\begin{aligned}W^2(\mathbf{y}|\mathbf{x}(D)) &= W(y_1|X_1)W(y_2|X_2) \\ &= \frac{1}{\pi^2} e^{-|y_1 - \sqrt{E_s} D|^2} e^{-|y_2 - \sqrt{E_s} D|^2}.\end{aligned}$$

The two likelihoods coincide when the exponents do, that is, when

$$|y_1 - \sqrt{E_s}|^2 + |y_2 - \sqrt{E_s}|^2 = |y_1|^2 + |y_2|^2.$$

Expanding $|y_i - \sqrt{E_s}|^2 = |y_i|^2 - 2\sqrt{E_s} \text{Re}(y_i) + E_s$, the terms $|y_i|^2$ cancel on both sides and the condition becomes $-2\sqrt{E_s}(\text{Re}(y_1) + \text{Re}(y_2)) + 2E_s = 0$, that is, $\text{Re}(y_1) + \text{Re}(y_2) = \sqrt{E_s}$. This is the boundary of the ML decoder: it decides $\hat{D} = 1$ when the sum of the real parts exceeds $\sqrt{E_s}$ and $\hat{D} = 0$ otherwise. Only the sum matters, since the two symbols carry

the same bit, and the imaginary parts play no role because both codewords are real.

7. A stream of bits interleaves control and information data so that 1 bit out of every 12 carries control commands. The stream is encoded with a channel code of rate $\frac{6}{7}$, and then groups of 7 coded bits are mapped onto sequences of n symbols of a 5-PAM constellation, with n the smallest length for which the mapping is reversible. What is the information rate of the system, in information bits (excluding control bits, before encoding) per modulation symbol?

- (a) $\frac{6}{5}$
- (b) $\frac{11}{5}$
- (c) $\frac{11}{8}$ *
- (d) 2

Solution: We follow 12 bits of the stream through the system. Of these, 11 are information bits and 1 is a control bit. The rate- $\frac{6}{7}$ code produces 7 coded bits for every 6 input bits, so the 12 bits become 14 coded bits, that is, two groups of 7. Each group of 7 bits has $2^7 = 128$ possible values, and n symbols of a 5-PAM constellation have 5^n values. Since $5^3 = 125 < 128$ and $5^4 = 625 \geq 128$, the shortest reversible mapping uses $n = 4$ symbols per group. The 12 stream bits thus occupy $2 \cdot 4 = 8$ symbols, and the information rate is

$$R = \frac{11 \text{ information bits}}{8 \text{ symbols}} = \frac{11}{8}$$

bits per symbol. The three stages each cost rate: the control bit ($\frac{1}{12}$), the code ($\frac{6}{7}$) and the mapping ($\frac{7}{4}$ bits per symbol instead of $\log_2 5 \approx 2.32$), and the product $\frac{1}{12} \cdot \frac{6}{7} \cdot \frac{7}{4}$ recovers $\frac{11}{8}$.